# COASTAL LANDFORMS / INDIA COAST AND CRZ

**COASTAL-SYSTEM BALANCE → REFRACTION MAP → EROSIONAL SEQUENCE → CLIFF RETREAT SECTION → DEPOSITIONAL DISCRIMINATOR → RELATIVE SEA-LEVEL FORK**

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

◆ PRIMARY CORE FLOW ● SUBORDINATE EXTRA ◆ APPROVAL / REVIEW

CORE BEFORE EXTRA • prior artifacts unchanged • exact same master drives poster and tiles

Approval: FALSE • Pending user review • source generation g4 and all prior artifacts unchanged

00

## STAGE 00 COASTAL-SYSTEM BALANCE

COASTAL-SYSTEM BALANCE | WAVES + TIDES + CURRENTS | MARINE ENERGY | RIVERS + CLIFFS + OFFSHORE | SEDIMENT SUPPLY | SEA LEVEL + LAND MOTION | ACCOMMODATION SPACE | HUMAN WORKS

WAVES + TIDES + CURRENTS → marine energy → RIVERS + CLIFFS + OFFSHORE → sediment supply → SEA LEVEL + LAND MOTION → accommodation space → HUMAN WORKS → redistribute erosion and accretion → MUST REMEMBER: Coastal morphology connects wave generation, refraction, erosion, longshore drift, sediment cells, deposition and

**CORE CLAIM**
- WAVES + TIDES + CURRENTS → marine energy
- RIVERS + CLIFFS + OFFSHORE → sediment supply

**MECHANISM**
- SEA LEVEL + LAND MOTION → accommodation space
- HUMAN WORKS → redistribute erosion and accretion

**CONSEQUENCE / LIMIT**
- MUST REMEMBER: Coastal morphology connects wave generation, refraction, erosion, longshore drift, sediment cells, deposition and
- sea-level/tectonic change to cliffs, caves-arches-stacks, beaches, spits, bars, lagoons, deltas and India's east-west coast contrast.

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MUST REMEMBER: Coastal morphology connects wave generation, refraction, erosion, longshore drift, sediment cells, deposition and sea-level/tectonic change to cliffs, caves-arches-stacks, beaches, spits, bars, lagoons, deltas and India's east-west coast contrast.**

01

## STAGE 01 REFRACTION MAP

REFRACTION MAP | DEEP WATER | WAVE FRONTS TRAVEL FASTER | SHALLOW HEADLAND | FRONTS SLOW AND CONVERGE | FRONTS DIVERGE AND ENERGY DISPERSES | EROSION FOCUS | DEPOSITIONAL SHELTER

DEEP WATER → wave fronts travel faster

SHALLOW HEADLAND → fronts slow and converge → BAY → fronts diverge and energy disperses → RESULT → erosion focus / depositional shelter

**CORE CLAIM**
- DEEP WATER → wave fronts travel faster
- SHALLOW HEADLAND → fronts slow and converge

**MECHANISM**
- BAY → fronts diverge and energy disperses
- RESULT → erosion focus / depositional shelter

**CONSEQUENCE / LIMIT**
- DEEP WATER → wave fronts travel faster
- SHALLOW HEADLAND → fronts slow and converge

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BAY → fronts diverge and energy disperses; RESULT → erosion focus / depositional shelter.**

02

## STAGE 02 EROSIONAL SEQUENCE

EROSIONAL SEQUENCE | HYDRAULIC ACTION + ABRASION | WIDENING AND ROOF ATTACK | ROOF COLLAPSE

JOINT / FAULT → hydraulic action + abrasion → CAVE → widening and roof attack → ARCH → roof collapse → STACK → STACK → undercutting → STUMP

**CORE CLAIM**
- JOINT / FAULT → hydraulic action + abrasion
- CAVE → widening and roof attack

**MECHANISM**
- ARCH → roof collapse → STACK
- STACK → undercutting → STUMP

**CONSEQUENCE / LIMIT**
- JOINT / FAULT → hydraulic action + abrasion
- CAVE → widening and roof attack

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: JOINT / FAULT → hydraulic action + abrasion.**

03

## STAGE 03 CLIFF RETREAT SECTION

CLIFF RETREAT SECTION | WAVE NOTCH | BASAL UNDERCUTTING | INSTABILITY AND COLLAPSE | DEBRIS REMOVAL | RENEWED ATTACK | WIDENING WAVE-CUT PLATFORM

**CORE CLAIM**
- WAVE NOTCH → basal undercutting
- OVERHANG → instability and collapse

**MECHANISM**
- DEBRIS REMOVAL → renewed attack
- RETREAT → widening wave-cut platform

**CONSEQUENCE / LIMIT**
- WAVE NOTCH → basal undercutting
- OVERHANG → instability and collapse

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: DEBRIS REMOVAL → renewed attack; RETREAT → widening wave-cut platform.**

04

## STAGE 04 DEPOSITIONAL DISCRIMINATOR

DEPOSITIONAL DISCRIMINATOR | ONE END ATTACHED | CROSSES BAY OR MOUTH | WATER ENCLOSED BEHIND BAR | JOINS ISLAND TO LAND

| CORE CLAIM | MECHANISM | CONSEQUENCE / LIMIT |
|---|---|---|
| SPIT → one end attached | LAGOON → water enclosed behind bar | SPIT → one end attached |
| BAR → crosses bay or mouth | TOMBOLO → joins island to land | BAR → crosses bay or mouth |

**CORE CLAIM**
- SPIT → one end attached
- BAR → crosses bay or mouth

**MECHANISM**
- LAGOON → water enclosed behind bar
- TOMBOLO → joins island to land

**CONSEQUENCE / LIMIT**
- SPIT → one end attached
- BAR → crosses bay or mouth

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: LAGOON → water enclosed behind bar; TOMBOLO → joins island to land.**

05

## STAGE 05 RELATIVE SEA-LEVEL FORK

RELATIVE SEA-LEVEL FORK | LAND RISE | SEA FALL | RAISED BEACH + TERRACE | EXPOSED MARINE FORMS | LAND FALL | SEA RISE | DROWNED RIVER VALLEY

LAND RISE / SEA FALL → emergence

RAISED BEACH + TERRACE → exposed marine forms → LAND FALL / SEA RISE → submergence → RIA → drowned river valley / FJORD → glacial trough

**CORE CLAIM**
- LAND RISE / SEA FALL → emergence
- RAISED BEACH + TERRACE → exposed marine forms

**MECHANISM**
- LAND FALL / SEA RISE → submergence
- RIA = drowned river valley / FJORD = glacial trough

**CONSEQUENCE / LIMIT**
- LAND RISE / SEA FALL → emergence
- RAISED BEACH + TERRACE → exposed marine forms

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RIA = drowned river valley / FJORD = glacial trough.**

06

## STAGE 06 STORM-SURGE CHAIN

STORM-SURGE CHAIN | CYCLONE WIND + LOW PRESSURE | WATER SETUP | TRACK + SHELF + BATHYMETRY | LOCAL AMPLIFICATION | HIGH TIDE + WAVES + RIVER FLOW | COMPOUND LEVEL | EXPOSURE + WEAK EVACUATION

CYCLONE WIND + LOW PRESSURE → water setup → TRACK + SHELF + BATHYMETRY → local amplification → HIGH TIDE + WAVES + RIVER FLOW → compound level → EXPOSURE + WEAK EVACUATION → disaster risk

**CORE CLAIM**
- CYCLONE WIND + LOW PRESSURE → water setup
- TRACK + SHELF + BATHYMETRY → local amplification

**MECHANISM**
- HIGH TIDE + WAVES + RIVER FLOW → compound level
- EXPOSURE + WEAK EVACUATION → disaster risk

**CONSEQUENCE / LIMIT**
- CYCLONE WIND + LOW PRESSURE → water setup
- TRACK + SHELF + BATHYMETRY → local amplification

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: HIGH TIDE + WAVES + RIVER FLOW → compound level.**

07

## STAGE 07 INDIA COAST CONTRAST

INDIA COAST CONTRAST | BROADER PLAINS + LARGE DELTAS | NARROWER PLAIN + MANY ESTUARIES | PORT TEST | DEPTH, SHELTER, SILTATION, ENGINEERING | TENDENCY, NOT UNIVERSAL BINARY

| CORE CLAIM | MECHANISM | CONSEQUENCE / LIMIT |
|---|---|---|
| EAST → broader plains + large deltas | PORT TEST → depth, shelter, siltation, engineering | EAST → broader plains + large deltas |
| WEST → narrower plain + many estuaries | RULE → tendency, not universal binary | WEST → narrower plain + many estuaries |

**CORE CLAIM**
- EAST → broader plains + large deltas
- WEST → narrower plain + many estuaries

**MECHANISM**
- PORT TEST → depth, shelter, siltation, engineering
- RULE → tendency, not universal binary

**CONSEQUENCE / LIMIT**
- EAST → broader plains + large deltas
- WEST → narrower plain + many estuaries

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PORT TEST → depth, shelter, siltation, engineering; RULE → tendency, not universal binary.**

08

## STAGE 08 NCCR EVIDENCE BOUNDARY

NCCR EVIDENCE BOUNDARY | 1990-2018 ASSESSMENT | ABOUT 33.6 PERCENT | ABOUT 26.9 PERCENT | ABOUT 39.6 PERCENT; NOT AN ANNUAL RATE

| CORE CLAIM | MECHANISM | CONSEQUENCE / LIMIT |
|---|---|---|
| PERIOD → 1990-2018 assessment | ACCRETING → about 26.9 percent | PERIOD → 1990-2018 assessment |
| ERODING → about 33.6 percent | STABLE → about 39.6 percent; not an annual rate | ERODING → about 33.6 percent |

**CORE CLAIM**
- PERIOD → 1990-2018 assessment
- ERODING → about 33.6 percent

**MECHANISM**
- ACCRETING → about 26.9 percent
- STABLE → about 39.6 percent; not an annual rate

**CONSEQUENCE / LIMIT**
- PERIOD → 1990-2018 assessment
- ERODING → about 33.6 percent

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: STABLE → about 39.6 percent; not an annual rate.**

09

## STAGE 09 CRZ IMPLEMENTATION LADDER

CRZ IMPLEMENTATION LADDER | CATEGORY-SPECIFIC NATIONAL RULES | HTL/LTL AND TECHNICAL INPUTS | CONSULTATION + APPROVAL + LOCAL MAP | APPRAISAL, DECISION, COMPLIANCE | CLOSE DISTINCTION | EMERGENT AND SUBMERGENT COASTS DESCRIBE RELATIVE SEA-LEVEL HISTORIES, EROSIONAL

NOTIFICATION → category-specific national rules

MAPPING → HTL/LTL and technical inputs → CZMP → consultation + approval + local map → PROJECT → appraisal, decision, compliance

**CORE CLAIM**
- NOTIFICATION → category-specific national rules
- MAPPING → HTL/LTL and technical inputs

**MECHANISM**
- CZMP → consultation + approval + local map
- PROJECT → appraisal, decision, compliance

**CONSEQUENCE / LIMIT**
- CLOSE DISTINCTION: Emergent and submergent coasts describe relative sea-level histories, erosional and depositional forms
- can coexist, and CRZ categories/regulatory lines are legal-planning constructs rather than geomorphic shoreline types.

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Emergent and submergent coasts describe relative sea-level histories, erosional and depositional forms can coexist, and CRZ categories/regulatory lines are legal-planning constructs rather than geomorphic shoreline types.**

10

## STAGE 10 COASTAL RESPONSE HIERARCHY

COASTAL RESPONSE HIERARCHY | KEEP NEW EXPOSURE OUT OF HIGH-RISK SPACE | BUFFERS + RESILIENT USE | MOVE ASSETS WHERE PERSISTENCE FAILS | SELECTIVE WORKS WITH DOWNDRIFT AUDIT | EVIDENCE LIMIT | EROSION CONTROL CAN SHIFT RISK DOWNDRIFT; PORTS, DAMS, SAND

AVOID → keep new exposure out of high-risk space

RESTORE / ACCOMMODATE → buffers + resilient use → RETREAT → move assets where persistence fails → PROTECT → selective works with downdrift audit

**CORE CLAIM**
- AVOID → keep new exposure out of high-risk space
- RESTORE / ACCOMMODATE → buffers + resilient use

**MECHANISM**
- RETREAT → move assets where persistence fails
- PROTECT → selective works with downdrift audit

**CONSEQUENCE / LIMIT**
- EVIDENCE LIMIT: Erosion control can shift risk downdrift
- ports, dams, sand mining, storms, subsidence and sea-level rise interact, while CRZ rules, clearances and project status require

MECHANISM / VERDICT — EVIDENCE LIMIT: Erosion control can shift risk downdrift; ports, dams, sand mining, storms, subsidence and sea-level rise interact, while CRZ rules, clearances and project status require exact notification/date/source discipline.

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Erosion control can shift risk downdrift; ports, dams, sand mining, storms, subsidence and sea-level rise interact, while CRZ rules, clearances and project status require exact notification/date/source discipline.**

11

## STAGE 11 COAST ANSWER SPINE

COAST ANSWER SPINE | DEFINE + MAP | PROCESS AND COASTAL SECTOR | ENERGY, SEDIMENT AND SEA-LEVEL CHAIN | PERIOD, CATEGORY, PLAN AND STATUS | INTEGRATED, ADAPTIVE AND LOCALLY MAPPED

DEFINE + MAP → process and coastal sector → DERIVE → energy, sediment and sea-level chain → VERIFY → period, category, plan and status → VERDICT → integrated, adaptive and locally mapped

**CORE CLAIM**
- DEFINE + MAP → process and coastal sector
- DERIVE → energy, sediment and sea-level chain

**MECHANISM**
- VERIFY → period, category, plan and status
- VERDICT → integrated, adaptive and locally mapped

**CONSEQUENCE / LIMIT**
- DEFINE + MAP → process and coastal sector
- DERIVE → energy, sediment and sea-level chain

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: DEFINE + MAP → process and coastal sector.**

E

## EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT | D.R. KHULLAR LISTS CYCLONE DANGERS AS GALES/STRONG WINDS, TORRENTIAL | KHULLAR NOTES MOST CASUALTIES FROM CYCLONES ARE CAUSED BY | TIDAL WAVES/STORM SURGES, WITH SEVERE SURGES PENETRATING INLAND FROM | LOW PRESSURE RAISES SEA LEVEL SLIGHTLY, BUT THE MAIN | IS STRONG ONSHORE CYCLONE WINDS PILING SEAWATER AGAINST A | INDIA LINK | THE SHALLOW, FUNNEL-SHAPED BAY OF BENGAL AND DELTAIC EAST

**OPTIONAL SOURCE DEPTH**
- D.R. Khullar lists cyclone dangers as gales/strong winds, torrential rain and high tidal waves, also called storm surges .
- Khullar notes most casualties from cyclones are caused by coastal inundation by
- tidal waves/storm surges, with severe surges penetrating inland from the coast.
- CAUTION: Mechanism: low pressure raises sea level slightly, but the main driver

**ADVANCED DEBATE**
- is strong onshore cyclone winds piling seawater against a shallow coast.
- CAUTION: India link: the shallow, funnel-shaped Bay of Bengal and deltaic east coast make Odisha-West Bengal-Bangladesh especially surge-prone.
- Majid Husain describes tropical muddy coasts with mangroves in sediment-rich lagoons, bays, deltas and estuaries, and also mentions salt marshes.
- D.R. Khullar notes mangroves occur along India's sheltered estuaries, tidal creeks, backwaters, salt marshes and mudflats.

**LEGACY / NUANCE**
- CAUTION: Blue carbon = carbon stored by coastal and marine ecosystems,
- especially mangroves, seagrasses and salt marshes, in biomass and waterlogged sediments.
- CAUTION: India link: Sundarbans, Mahanadi- Godavari- Krishna deltas, Gulf of Kachchh, Gulf of
- Mannar and Andaman-Nicobar ecosystems link coastal protection with climate mitigation.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels